CHAPTER 6

THE SOLUTION OF NONLINEAR EQUATIONS

6.1 INTRODUCTION

Suppose we wish to find a solution of an equation of the form

$$f(x) = 0.$$

In this chapter we shall discuss some methods for finding an approximate solution of the above equation in the following sense. We say that ξ is an approximate solution of the equation $f(x) = 0$ if either $|f(\xi)| < \epsilon_1$ or $|\xi - r| < \epsilon_2$. Here r is an exact (unknown) solution of the equation and ϵ_1 and ϵ_2 are fixed positive numbers chosen according to the required accuracy.

The trigonometric, exponential and logarithmic functions are examples of functions called transcendental functions. If f(x) is a transcendental function then the equation $f(x) = 0$ is called a transcendental equation. For example, $\tan x - x = 0$ is a transcendental equation.

DEFINITION 6.1: For a given real-valued function, a number r such that $f(r)=0$ is called a **root** of equation $f(x)=0$ or a **zero** of the function f(x). For example, if $f(x)=(x-1)e^x$, then $x = 1$ is a root of the equation $f(x) = 0$ as $f(1) = 0$.

DEFINITION 6.2: For a function f(x) which is sufficiently differentiable, a root of multiplicity m is a number r such that $f(r) = f'(r) = f''(r)=\ldots=f^{m-1}(r)=0$ but $f^m(r) \neq 0$. For example, if $f(x) = (x-1)e^x$ then $f'(x)=xe^x$. As $f(1)=0$ but $f'(1) \neq 0$, therefore $x = 1$ is a root of multiplicity 1 or a simple root of the equation $f(x) = 0$. However, when $f(x) = (x - 1)^2 e^x$ then

$$f'(x) = (x + 1)(x - 1)e^x$$

$$f''(x) = (x^2 + 2x - 1)e^x.$$

Since $f(1) = 0$, $f'(1) = 0$ and $f''(1) = 2e \neq 0$,

hence $x = 1$ is a root of multiplicity 2 of the equation $f(x) = 0$.

6.2 BISECTION METHOD

Let f be continuous on an interval [a,b]. If f(a) and f(b) have opposite signs then the function f must take all values between f(a) and f(b), therefore it must vanish at least once between a and b, Fig. 6.1. This means that

$$a < r < b \text{ , where } f(r) = 0.$$

Now if we take ξ_1 any point in (a,b) then

$$|\xi_1 - r| < b - a.$$

To get closer to the root r we evaluate the function at the mid point $\frac{a+b}{2}$ of (a,b).

Now if $f(a)\, f(\frac{a+b}{2}) \leq 0$ then $r \in (a, \frac{a+b}{2}]$

and if $f(b)\, f(\frac{a+b}{2}) \leq 0$ then $r \in [\frac{a+b}{2}, b)$.

If, for example, the second possibility happens to be true and we take ξ_2 any point in $(\frac{a+b}{2}, b)$ then we know that

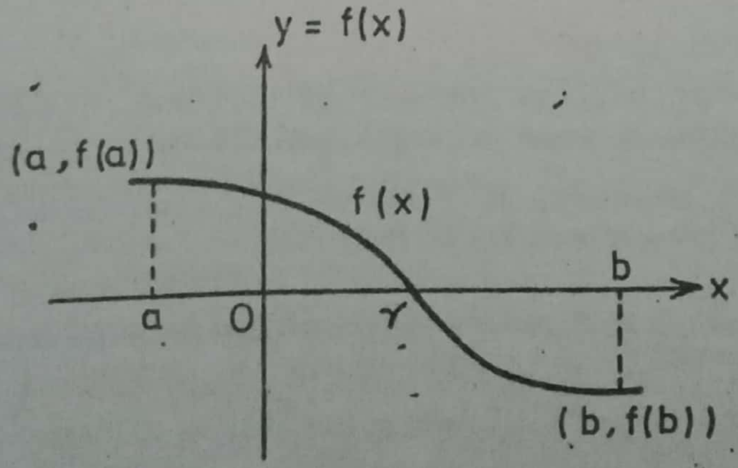

Fig. 6.1 Bisection method.

$$|\xi_2 - r| < \frac{b - a}{2}.$$

We can continue this procedure and after every step we locate the root in an interval whose width is half that of the interval in the previous step. For this reason this method is known as the **bisection method** or the **method of interval halving**. We illustrate the method by an example.

EXAMPLE 6.1: Find a root of the equation

$$2x^3 + x - 2 = 0. \tag{6.1}$$

SOLUTION: Let $f(x) = 2x^3 + x - 2$.

It is easily seen that $f(0) = -2$ and $f(1)=1$. Thus $f(0)$ and $f(1)$ have opposite signs and a root r of the equation must lie in $(0,1)$. We can take $\xi_1=(0+1)/2 =0.5$ with absolute error $|\xi_1-r| \le 1$. We evaluate $f(x)$ at this point. We see that $f(0.5) = -1.25$.

Since $\qquad f(0.5)\ f(1) < 0,$

we conclude $\qquad 0.5 < r < 1,$

and we can take $\xi_2 = 0.75$ with absolute error i.e. $|\xi_2-r| \le 0.50$. Next we evaluate f at 0.75. Now $f(0.75) = -0.40625$. Thus we see that

$$0.75 < r < 1$$

and if we take $\xi_3 = \frac{1+0.75}{2} = 0.875$, then

$$|\xi_3 - r| \le 0.25.$$

We continue in this manner and find

$$f(0.875) = 0.2148. \text{ Thus } 0.75 < r < 0.875,$$

and if we take $\xi_4 = \frac{0.75+0.875}{2} = 0.8125$, then $|\xi_4 - r| \le 0.125$.

Now $\quad f(\xi_4) = -0.1147$, hence $0.8125 < r < 0.875$

and $\xi_5 = 0.84375$ is such that $|\xi_5 - r| \leq 0.0625$.

Thus ξ_5 is correct to the first digit after the decimal point.
After seven more steps we find that

$$\xi_{12} = 0.8352055, \; f(\xi_{12}) = -0.000431$$

and

$$0.83521 < r < 0.83545.$$

The last line indicates that $\xi = 0.835$ is an approximate root correct to three decimal positions. These values are tabulated below.

Table 6.1 Successive Bisection Iterations

n	a_n	ξ_n	b_n	$f(a_n)$	$f(\xi_n)$	$f(b_n)$
1	0.0	0.5	1.0	-2.0	-1.25	1.0
2	0.5	0.75	1.0	-1.25	-0.40625	1.0
3	0.75	0.875	1.0	-0.40625	0.2148	1.0
4	0.75	0.8125	0.875	-0.40625	-0.1147	0.2148
5	0.8125	0.84375	0.875	-0.1147	0.0451	0.2148
.		.			.	
.		.			.	
.		.			.	
12		0.8352055			-0.000431	

The bisection method generates a sequence $\xi_1, \xi_2, \ldots$ which does converge to the exact root. This method suffers from the disadvantage that the convergence is slow but it has the advantages that the iteration always converges and at every step we have available an estimate of accuracy.

Now $f(\xi_4) = -0.1147$, hence $0.8125 < r < 0.875$

and $\xi_5 = 0.84375$ is such that $|\xi_5 - r| \le 0.0625$.

Thus ξ_5 is correct to the first digit after the decimal point. After seven more steps we find that

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4	0.75	0.8125	0.875	-0.40625	-0.1147	0.2148
5	0.8125	0.84375	0.875	-0.1147	0.0451	0.2148
.		.			.	
.		.			.	
.		.			.	
12		0.8352055			-0.000431	

The bisection method generates a sequence ξ_1, ξ_2, ... which does converge to the exact root. This method suffers from the disadvantage that the convergence is slow but it has the advantages that the iteration always converges and at every step we have available an estimate of accuracy.

$m = \frac{1+2}{2}$

$m = 1.5$

1, 2

0.12

EXERCISE 6.1

1. Use the bisection method to find solutions correct to three decimal places for the following:

   a) $f(x) = xe^{x} - 5 = 0$, for $1 \le x \le 2$,

   b) $f(x) = 2e^{-x} - \sin x = 0$, for $0 \le x \le 1$,

   c) $f(x) = x - e^{1/x} = 0$, for $1 \le x \le 2$.

   $f(x)$
   $f'(x)$
   x

2. The function $f(x)= x^3+4x^2-10$ has a root in [1,2]. Approximate the root correct to 3 decimal positions by bisection method.

3. Find an approximation to $\sqrt[3]{25}$ correct to three significant digits, using the bisection method. [Hint, $x^3 - 25 = 0$]

4. Aproximate the solution of $x^3 - x - 1 = 0$, lying in the interval [1,2] with accuracy 10^{-2}.

5. Approximate with accuracy 10^{-3} to the solution of $x^3+x-4 = 0$, lying in the interval [1,4].

6. Find an interval within which a solution of the following equations is located. Then solve the equations by bisection method, obtaining the solution to within 0.1%.

   a) $x^4 + 2/x^4 = 1000$,

   b) $x \sin x = 50$,

   c) $x + x^x = 100$.

   $x^2 - 3 = 0$

   $x^2 - 3 = 0$

7. Find an approximation to $\sqrt{3}$ correct to four significant digits using the bisection method.

   $x = \sqrt{3}$
   $x^2 = 3$

   $x^2 - 3$

6.3 THE METHOD OF FALSE POSITION OR REGULA FALSI METHOD

In the bisection method, we form a sequence $(\xi_r)_{r=1}^{\infty}$ which converges to the exact root 'r' of the equation $f(x)=0$. The point ξ_r is chosen as the mid point of an interval in which the root is located. A modification of this method is to choose ξ_1 as the point where the chord joining the points $(a,f(a))$ and $(b,f(b))$ intersects the x-axis. In Fig.6.2, ξ_1 is seen to be much closer to r as compared with the point $\frac{a+b}{2}$. Next we compare the signs of f(a) and f(b) with that of $f(\xi_1)$. In Fig.6.2, $f(b)\ f(\xi_1) < 0$ and we choose ξ_2 as the point where the chord joining $(\xi_1, f(\xi_1))$ and (b,f(b)) intersects the x-axis. The sequence of points ξ_1, ξ_2, ... generated in this manner converges much faster than the corresponding sequence in the bisection method. Now the equation of the chord $\overline{PQ}$ is

$$y - f(b) = \frac{f(b) - f(a)}{b - a}(x - b)$$

This line intersects the x-axis at the point $(\xi_1, 0)$, hence

$$-f(b) = \frac{f(b) - f(a)}{b - a}(\xi_1 - b)$$

or

$$\xi_1 = -\frac{(b - a)f(b)}{f(b) - f(a)} + b$$

$$= \frac{a\ f(b) - b\ f(a)}{f(b) - f(a)}. \qquad (6.2)$$

We use the above formula to find ξ_r at each step.

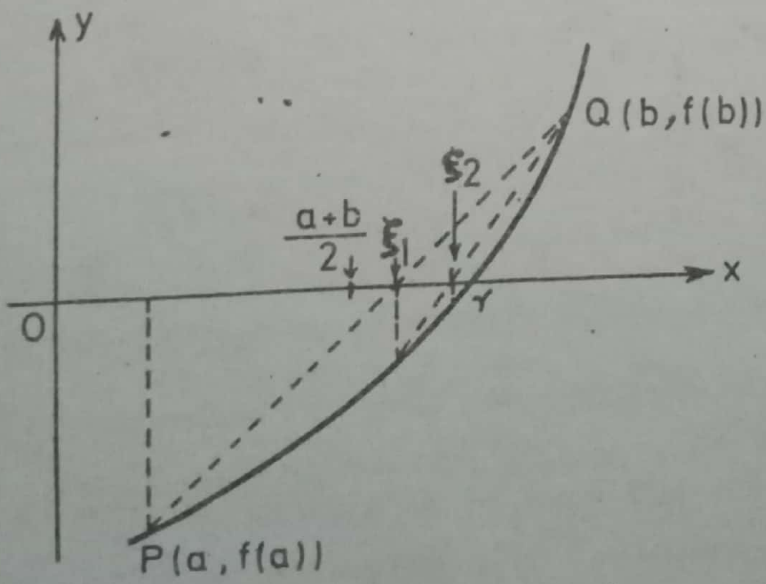

Fig. 6.2 Regula Falsi method.

We illustrate the method by solving the same example as we did in the last section.

EXAMPLE 6.2: Solve $2x^3 + x - 2 = 0$ by regula falsi method.

SOLUTION: Here $f(x)=2x^3+x-2$. Let $a=0$, $b=1$, then $f(0)=-2$, $f(1)=1$, and $f(0)\,f(1) < 0$,

$$\xi_1 = \frac{a.f(b) - b.f(a)}{f(b) - f(a)} = \frac{0(1)-1(-2)}{1-(-2)} = 0.66667.$$

Now $f(\xi_1) = -0.74073$ and $f(\xi_1)\,f(1) < 0$, so

$$\xi_2 = \frac{\xi_1 f(1) - 1\, f(\xi_1)}{f(1) - f(\xi_1)} = 0.80851 \text{ with } |\xi_2 - r| \le 0.33333.$$

We find $f(\xi_2) = -0.13446$ and $f(\xi_2)\,f(1) < 0$.

Thus $$\xi_3 = \frac{\xi_2 f(1) - 1\, f(\xi_2)}{f(1) - f(\xi_2)} = 0.83121.$$

Now $f(\xi_3) = -0.02022$ and $f(\xi_3)\,f(1) < 0$.

$$\xi_4 = \frac{\xi_3 f(1) - 1\, f(\xi_3)}{f(1) - f(\xi_3)} = 0.83455.$$

Table 6.2 Successive Regula Falsi Iterates

n	ξ_n	$f(\xi_n)$
1	0.66667	-0.74073
2	0.80851	-0.13446
3	0.83121	-0.02022
4	0.83455	-0.00297

If we round-off to three decimal positions the above result is 0.835 which is correct to 3 figures. Thus we see that the sequence ξ_1, ξ_2, ... converges to r faster but at each step we have to perform more calculations.

10=2

EXERCISE 6.2

1. Find the roots of the following equations by regula falsi method correct to 4 significant figures:

   a) $x^3 - 3x + 1 = 0$, for $1 \le x \le 2$

   b) $e^x - 3x = 0$, for $0 \le x \le 1$

   c) $e^x = 10 - x$, for $2 \le x \le 3$.

2. Find a root of the equation

$$x^3 - 4x^2 + x - 10 = 0$$

by regula falsi method correct to 3 decimal positions taking $x_1 = 4$, $x_2 = 5$.

3. Find a real solution of the equation

$$f(x) = x^3 - 5x^2 - 29 = 0$$

using regula falsi method correct to 4 significant figures taking $x_1 = 5$ and $x_2 = 6$.

4. Apply the regula falsi method to find an approximate root of $\cos x - x = 0$, correct to 6 decimal positions lying in the interval $[.5, \pi/4]$.

5. Approximate the solutions to within 10^{-5} of the following equations using regula falsi method.

   a) $3x^2 - e^x = 0$, b) $x^2 + 10 \cos x = 0$.

6.4 SECANT METHOD

In this method we simply calculate ξ_{r+1} from the formula (with $a = \xi_0$, $b = \xi_1$)

$$\xi_{r+1} = \frac{\xi_{r-1} f(\xi_r) - \xi_r f(\xi_{r-1})}{f(\xi_r) - f(\xi_{r-1})}, \quad r = 1, 2, \ldots \tag{6.3}$$

or

$$\xi_{r+1} = \xi_r - \frac{(\xi_r - \xi_{r-1}) f(\xi_r)}{f(\xi_r) - f(\xi_{r-1})}$$

and give up the test to locate the root in smaller intervals. If we apply this method to the example we have already discussed, we obtain

$$\xi_2 = \frac{\xi_0 f(\xi_1) - \xi_1 f(\xi_0)}{f(\xi_1) - f(\xi_0)} = \frac{0\times f(1) - 1\times f(0)}{f(1) - f(0)} = 0.66667.$$

$$\xi_3 = \frac{\xi_1 f(\xi_2) - \xi_2 f(\xi_1)}{f(\xi_2) - f(\xi_1)} = \frac{1\times f(0.66667) - 0.66667\times f(1)}{f(0.66667) - f(1)}$$

$$= \frac{1(-0.74073) - 0.66667\times 1}{-0.74073 - 1} = 0.80851.$$

$$\xi_4 = \frac{\xi_2 f(\xi_3) - \xi_3 f(\xi_2)}{f(\xi_3) - f(\xi_2)}$$

$$= \frac{0.66667\times f(0.80851) - 0.80851\times f(0.66667)}{f(0.80851) - f(0.66667)}$$

$$= \frac{0.66667(-0.13446) - 0.80851(-0.74073)}{-0.13446 - (-0.74073)} = 0.83997.$$

$$\xi_5 = \frac{\xi_3 f(\xi_4) - \xi_4 f(\xi_3)}{f(\xi_4) - f(\xi_3)}$$

$$\xi_5 = \frac{0.80851\times f(0.83997) - 0.83997\times f(0.80851)}{f(0.83997) - f(0.80851)}$$

$$= \frac{0.80851(0.02525) - 0.83997(-0.13446)}{0.02525 + 0.13446} = 0.83495.$$

$$\xi_6 = \frac{\xi_4 f(\xi_5) - \xi_5 f(\xi_4)}{f(\xi_5) - f(\xi_4)}$$

$$= \frac{0.83997\times f(0.83495) - 0.83495\times f(0.83997)}{f(0.83495) - f(0.83997)}$$

$$= \frac{0.83997(-0.00089) - 0.83495(0.02525)}{-0.00089 - 0.02525} = 0.83512.$$

Tabulating the results in the table 6.3.

Table 6.3 Successive Secant Iterations

n	ξ_n	$f(\xi_n)$
0	0.0	-2.0
1	1.0	1.0
2	0.66667	-0.74073
3	0.80851	-0.13446
4	0.83997	0.02525
5	0.83495	-0.00089
6	0.83512	-0.00001

EXERCISE 6.3

1. Find the smallest positive root of

$$f(x) = \cos x \cosh x - 1 = 0$$

correct to 3 decimal positions by secant method.

2. Find a root of $f(x)=\cos x-x=0$, correct to 5 decimal places by secant method.

3. Approximate the solutions to within 10^{-5} of the following equations using secant method:

a) $3x^2 - e^x = 0$ b) $x^2 + 10 \cos x = 0$.

4. Find the solutions correct to 4 significant figures of the following equations using secant method:

a) $e^x = 10 - x$ b) $x^3 - 5x^2 - 29 = 0$ c) $x^x = 10$.

6.5 NEWTON-RAPHSON METHOD

Let (x_0,y_0) be a point on the curve $y = f(x)$, Fig.6.3. Draw the tangent line to the curve at (x_0,y_0). The equation of the tangent line is

$$y-y_0 = f'(x_0)(x - x_0).$$

If this line intersects the x-axis at the point $x = x_1$ then

$$x_1 = x_0 - \frac{y_0}{f'(x_0)}$$

$$= x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Thus if x_0 is an approximation to the root then x_1 will in general, be a better approximation and we can generate a sequence $x_0, x_1, \ldots$ by using the iteration formula

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad k = 0, 1, \ldots \tag{6.4}$$

If $f'(x)$ can be easily calculated and does not vanish at r, the sequence $\{x_k\}_{k=0}^{\infty}$ converges much faster than any of the methods discused above. For this reason this method, known as Newton-Raphson method, is very popular.

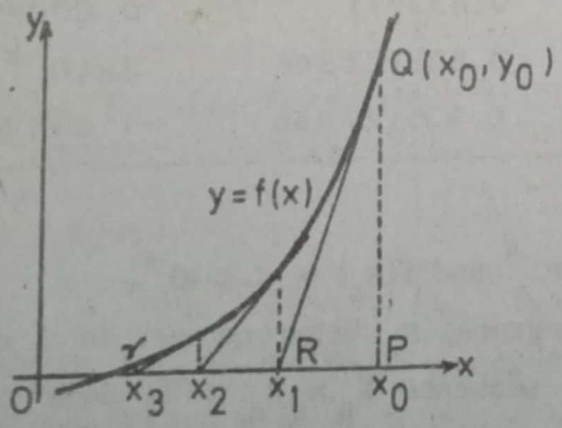

Fig. 6.3 Newton-Raphson method

EXAMPLE 6.3: Find a root of $2x^3+x-2 = 0$ by Newton-Raphson method.

SOLUTION: Let $f(x) = 2x^3+x-2$, then $f'(x) = 6x^2+1$. Take $x_0 = 1$.

$$x_1 = 1 - \frac{1}{7} = \frac{6}{7} = 0.856$$

$$x_2 = 0.856 - \frac{0.11044}{5.39642} = 0.83553$$

$$x_3 = 0.83553 - \frac{0.00211433}{5.8867727} = 0.835171$$

$$x_4 = 0.835171 - \frac{0.00025225}{5.1850636} = 0.835122348$$

$$x_5 = 0.835122348 - \frac{0.000000008}{5.18457604} = 0.835122346.$$

These successive iterations are given in Table 6.4.

Table 6.4 Successive Newton-Raphson Iterations

n	x_n	$f(x_n)$
0	1.0	1.0
1	0.856	0.11044032
2	0.83553	0.002114333
3	0.835171	0.000252249
4	0.835122348	2×10^{-9}
5	0.835122346	-1.2×10^{-8}

Notice that $f(x_4) = 2\times10^{-9}$ and $f(x_5) = -1.2\times10^{-8}$.

Notice the fast convergence, x_5 being correct to 9 decimal positions. In general the sequence $x_0, x_1, x_2, \ldots$ is such that if x_k is correct to a certain number of decimal positions then x_{k+1} is correct to two more positions. This fact is expressed by saying that the convergence of Newton-Raphson method is quadratic. We shall discuss this in more detail in section 6.7.

EXERCISE 6.4

1. Solve the following equations correct to 4 significant figures, using Newton-Raphson method.

   a) $x^3 - 5x^2 - 29 = 0$ b) $e^x = 10 - x$

   c) $e^{-x/4}(2 - x) = 1$ d) $6\theta = 5 \sinh \theta$

2. Using Newton-Raphson method, compute a negative root of the equation $f(x) = x^4 - 3x^2 + 75x - 10000 = 0$ correct to five significant figures.

3. Find the real roots of the following equations correct to 5 decimal positions using Newton-Raphson method.

   a) $x e^x = 1$, b) $x^2 + 4 \sin x = 0$, c) $3x - \cos x - 1 = 0$,

   d) $2x - 3 \sin x - 5 = 0$, e) $2x - \log_{10} x = 7$,

   f) $x \sinh(10/x) - 15 = 0$, g) $x \log_{10} x = -0.125$.

4. Find a positive root of the equation

   $$f(x) = x^3 - 0.2x^2 - 0.2x - 1.2 = 0$$

   to an accuracy of 0.002.

5. Find a solution of the equation $x = \cos x$ on the interval $[0, \pi/4]$ by Newton - Raphson method correct to 5 decimal positions.

6. The function $f(x) = (4x-7)/(x-2)$ has a zero $r = 1.75$. Use Newton-Raphson method with the following initial approximations.

   a) $x_0 = 1.625$ b) $x_0 = 1.875$

   c) $x_0 = 1.5$ d) $x_0 = 1.95$

7. Find by Newton-Raphson method the smallest positive root of

   $$x^x + 2x = 6$$

   to three decimal places.

8. Find the smallest positive root of the function

   $$f(x) = x^2 |\sin x| - 4$$

   by Newton - Raphson method.

6.6 FIXED POINT ITERATION

A point α is said to be a fixed point of a function g if $g(\alpha)=\alpha$. For example '0' is a fixed point of sin x, 0.7390851 is a fixed point (correct to 7 decimal positions) of cos x but the functions ln x and exp(x) have no fixed point.

Suppose we wish to find a root of the equation

$$f(x) = 0 \tag{6.5}$$

We rearrange this equation in the form

$$x = g(x) \tag{6.6}$$

Thus any fixed point of g will be a root of the equation (6.5). The function g is called an iteration function and to find a fixed point of g we start with an arbitrary point x_0 and find $x_1, x_2, \ldots$ from the formula

$$x_{n+1} = g(x_n) \tag{6.7}$$

If the sequence converges, it will do so to a fixed point of g, hence to a root of $f(x) = 0$.

The rearrangement of $f(x) = 0$ in the form $x = g(x)$ can be done in many ways. For example the equation

$$x^3 - 13x - 12 = 0 \tag{6.8}$$

which has -1,-3,4 as roots, may be written as

$$\text{(a)}\quad x = \sqrt{\frac{13x + 12}{x}},\quad g(x) = \sqrt{\frac{13x + 12}{x}}$$

$$\text{(b)}\quad x = \frac{x^3 - 12}{13},\qquad g(x) = \frac{x^3 - 12}{13},$$

$$\text{(c)}\quad x = \frac{12}{x^2 - 13},\qquad g(x) = \frac{12}{x^2 - 13},$$

$$\text{(d)}\quad x = x^3 - 12x - 12,\quad g(x) = x^3 - 12x - 12.$$

Now let us take $x_0 = 1$. The iteration $x_{n+1} = g(x_n)$ in (a) gives $x_1 = 2.236$, $x_2 = 4.286$, $x_3 = 3.975$, $x_4 = 4.00236$, $x_5 = 3.9998$. The sequence converges to the root 4 of the equation (6.8).

If we take $g(x) = \frac{x^3-12}{13}$ of (b) and start with $x_0 = 1$, we get $x_1=-0.846$, $x_2=-0.9697$, $x_3=-0.99321$, $x_4=-0.998444$ $x_5=-0.99964,\ldots.$ This sequence converges to the root -1 of equation (6.8). Now take $x_0 = 5$, then $x_1 = 8.692$, $x_2 = 49.597$, ... which diverges.

Now consider (c). Let us take $x_0 = 3.5$ which is 'close' to 4, a fixed point of $\frac{12}{x^2-13}$. The iteration $x_{n+1} = g(x_n)$ gives $x_1 = -16$, $x_2=0.0494$, $x_3=-0.92325$, $x_4=-0.98785$, $x_5=-0.99799$, $x_6=-0.99967,\ldots$ The sequence converges to -1. If we take $x_0 = 1$, the iteration $x_{n+1} = g(x_n)$ gives $x_1 = -1$, $x_2 = -1$, $x_3 = -1$, ...

Finally the iteration

$$x_{n+1} = x_n^3 - 12x_n - 12$$

which corresponds to the re-aranegment (d) gives the sequence

$$x_0 = 1,\ x_1 = -23,\ x_2 = -11903,\ x_3 = -1.686433\times10^{12},\ldots.$$

The sequence obviously diverges.

The above example illustrates the fact that the iteration $x_{n+1} = g(x_n)$ may or may not converge. Also an iteration function may yield a convergent sequence for a choice of x_0 but a divergent sequence for another choice of x_0. Again a choice of x_0 near a fixed point might yield a sequence which converges not to the desired point but to another point. We now consider a set of conditions for the existence of a fixed point and for the convergence of the iteration $x_{n+1} = g(x_n)$ to this point.

THEOREM 6.1: Let $g(x)$ be continuous on an interval $[a,b]$ and let $g(x)$ map $[a,b]$ into itself, then g has a fixed point in $[a,b]$.

PROOF: If $g(a) = a$ or $g(b) = b$ then the theorem is proved.

Suppose $g(a) \neq a$ and $g(b) \neq b$. Since $g(a) \in [a,b]$ and $g(b) \in [a,b]$ hence $g(a)-a > 0$ and $g(b)-b < 0$. Thus the function $g(x)-x$, which is continuous on $[a,b]$, has opposite signs at points $x=a$ and $x=b$. Therefore there exists a point ξ, $a < \xi < b$ so that $g(\xi) - \xi = 0$. This means ξ is a fixed point of $g(x)$.

THEOREM 6.2: Let $g(x)$ be such that

(a) g maps $[a,b]$ into itself
(b) g is differentiable on $[a,b]$ and $|g'(x)| \le K < 1$ for x in $[a,b]$, then
(i) g has a fixed point ξ in $[a,b]$,
(ii) the fixed point is unique,
and (iii) starting with any x_0 in $[a,b]$, the iteration

$$x_{n+1} = g(x_n) \text{ converges to } \xi.$$

PROOF: (i) Differentiability of g on $[a,b]$ implies continuity of g on that interval, hence by Theorem 6.1, a fixed point ξ exists in $[a,b]$.

(ii) If there are more than one, say two fixed points α and ξ in $[a,b]$ then, supposing $\xi < \alpha$

$$\alpha = g(\alpha)$$
$$\xi = g(\xi) \qquad (6.9a)$$

and

$$\alpha - \xi = g(\alpha) - g(\xi) \qquad (6.9b)$$
$$= g'(\eta)(\alpha - \xi),$$

η is a point in the interval $[\xi,\alpha]$. Eq. (6.9b) follows (6.9a) on applying the mean value theorem to the function $g(x)$ on the interval $[\xi,\alpha]$. From (6.9) we have

$$|\alpha - \xi| = |g'(\eta)|\ |\alpha - \xi| \le K\ |\alpha - \xi| < |\alpha - \xi|$$

which is impossible. Hence the fixed point must be unique.

(iii) To show that the sequence x_0, x_1, ... converges to ξ, note that

$$\xi = g\ (\xi)$$

and

$$x_n = g\ (x_{n-1})$$

Thus

$$\xi - x_n = g(\xi) - g(x_{n-1})$$
$$= g'(\eta)(\xi - x_{n-1}),\ \eta \in [\xi, x_{n-1}]$$

by the mean value theorem. Therefore

$$|\xi - x_n| = |g'(\eta)|\ |\xi - x_{n-1}| \le K\ |\xi - x_{n-1}|.$$

Similarly
$$|\xi - x_{n-1}| \le K\, |\xi - x_{n-2}|$$
$$\vdots$$
$$|\xi - x_1| \le K\, |\xi - x_0|.$$

From the above inequalities, we obtain

$$|\xi - x_n| \le K^n\, |\xi - x_0|$$

and

$$\lim_{n \to \infty} (\xi - x_n) \le \lim_{n \to \infty} K^n |\xi - x_0| = 0,$$

because $K < 1$. From the above result it follows that

$$\lim_{n \to \infty} |\xi - x_n| = 0,$$

or

$$\lim_{n \to \infty} x_n = \xi.$$

Thus the Theorem is proved.

Sometimes it is difficult to verify the assumption that g maps $[a,b]$ into itself. However if $|g'(\xi)| < 1$ and $g'(x)$ is continuous in an interval containing ξ it can be shown that if x_0 is taken close to ξ then the iteration.

$$x_{n+1} = g(x_n)$$

will converge to ξ.

EXAMPLE 6.4: Consider the example discussed in sections 6.2 to 6.5.

$$2x^3 + x - 2 = 0$$

If we re-arrange the equation and write

$$x = x - \frac{2x^3 + x - 2}{7}$$

Then
$$g(x) = x - \frac{2x^3 + x - 2}{7}$$

$$g'(x) = \frac{6 - 6x^2}{7} = \frac{6}{7}(1 - x^2)$$

and $|g'(x)| < 1$ if we take $|x| < 1$ and the iteration

$$x_{n+1} = g(x_n)$$

will converge to ξ if we take x_0 close to ξ which is known to lie in (0.5,1).

Thus if $x_0 = 1$

$$x_1 = 1 - \frac{1}{7} = 0.8556$$

$$x_2 = 0.8556 - \frac{0.108287}{7} = 0.84013$$

$$x_3 = 0.84013 - \frac{0.026091}{7} = 0.836403$$

$$x_4 = 0.836403 - \frac{0.006650}{7} = 0.835453$$

6.7 ORDER OF CONVERGENCE

Let us define

$$e_n = x_n - \xi$$

where ξ is an exact root of the equation $f(x) = 0$ and x_n is the n-th term of a sequence x_1, x_2, ... generated in some manner to approximate this root. When n is large and

$$|e_{n+1}| \approx K\, |e_n|^p$$

where K is a constant, then p is said to be the order of convergence of the method. If $p = 1$, the method is said to converge linearly. The secant method is such a method. For the regula falsi method, it can be shown Fröberg [4] that $p = 1.62$. We shall now show that, in general, for Newton's method $p = 2$. For this reason the method is said to converge quadratically. First we prove the following lemma.

LEMMA: Let f be twice differentiable at ξ, $f(\xi)= 0$ and $f'(\xi) \neq 0$, then the derivative of the iteration function for Newton's method i.e.

$$g(x) = x - f(x)/f'(x)$$

vanishes at ξ.

PROOF: Here $g(x) = x - \frac{f(x)}{f'(x)}$, then

$$g'(x) = 1 - \frac{[f'(x)]^2 - f(x)f''(x)}{[f'(x)]^2}$$

and $$g'(\xi) = 1 - \frac{[f'(\xi)]^2}{[f'(\xi)]^2} = 0, \text{ since } f(\xi) = 0.$$

Now we shall prove that Newton's method converges quadratically.

THEOREM 6.3: The order of convergence for Newton's method is 2.

PROOF: Define
$$\begin{aligned} e_{n+1} &= x_{n+1} - \xi \\ &= g(x_n) - g(\xi) \\ &= g(\xi + x_n - \xi)-g(\xi) \\ &= g(\xi) + (x_n - \xi)g'(\xi) + \frac{1}{2}(x_n-\xi)^2 g''(\eta) - g(\xi) \end{aligned}$$

where η is a point between ξ and x_n. Using the fact that $g'(\xi)=0$, and $e_n = x_n - \xi$, we have

$$e_{n+1} = \frac{1}{2} g''(\eta)e_n^2$$

Thus

$$|e_{n+1}| = K\,|e_n|^2$$

where $K = \frac{1}{2}|g''(\eta)|$ and is approximately a constant for large n. This proves the theorem.

In practical terms the theorem means that when a sequence $x_1, x_2, \ldots$ converges to ξ and x_r is correct to one decimal position, (i.e. $e_r \approx 10^{-1}$), the next approximate value x_{r+1} will be correct to two decimal positions ($e_{r+1} \approx 10^{-2}$), x_{r+2} to 4 decimal positions ($e_{r+2} \approx 10^{-4}$) and so on. It should also be remarked that if $f'(\xi)=0$, then ξ is a double root of the equation $f(x)= 0$.

In this case Newton's method no longer converges quadratically, the convergence is only linear.

EXERCISE 6.5

1. For each of the following equations, find an interval within which a solution is located. Then solve the equation by fixed point iteration accurate to within 10^{-2}. Check that the convergence criterion is satisfied for the particular arrangement of the equation used.

   a) $x \sin x = 50$, b) $x + x^x = 100$ $\quad x = \frac{\ln(100-x)}{\ln x}$

2. Use fixed point iteration to find an approximate solution of $x - 4^{-x} = 0$ accurate to within 10^{-2}.

3. For an equation $3x^2 - e^x = 0$ determine a function $g(x)$ and ~~an interval [a,b] such~~ that fixed point iteration will converge to a positive solution of the equation. Find the solution to within 10^{-3}. $\quad x = 0.25 + \sin x$

4. Find the real roots of the equation $x - \sin x = 0.25$ to three significant digits, using fixed point iteration.

   $x = 0.25 + \sin x$

5. Find the largest positive root r of the equation

   $$x^3 + x = 1000 \quad \Rightarrow \quad x^3 = 1000 - x$$

   to within 10^{-2}, using fixed point iteration. $\quad x = \sqrt[3]{1000 - x}$

6. Find the real roots of the equation

   $f(x) = x^3 - 3x + 1 = 0$, using fixed point iteration. $\quad \Rightarrow x^3 = \sqrt[3]{3x-1}$

7. Find a root near 1.5 of $x = 1/2 + \sin x$, using fixed point iteration.

8. Apply the fixed point iteration to the equation $x = e^{-x}$ to find a root near $x = 0.5$. Show that starting with $x_0 = 0.5$, the approximations x_{10} and x_{11} agree to three places at .567.

9. Find by fixed point iteration a real root of

   $$2x - \log_{10} x = 7.$$